ME 262 Theory of Machines I
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F. W. ADAM

MECHANICAL ENGINEERING DEPARTMENT

KNUST

Grading:

- Homework & Attendance 10%
- Midterm Exam 10%
- Project 10%
- Final Exam 70%

Prerequisite: A previous course in ME 162, MATH 151, MATH 152, MATH 251, MATH 252, ME 255.

Homework: About 5-10 problems will be assigned every other week and these will be collected for grading. Collaboration is encouraged, but independent work is required and essential in order to perform well on the exams.

REFERENCE BOOKS

- Mechanics of Machines, Elementary Theory and Examples, Hannah J. and Stephens R. C.
- Mechanism Design: Analysis and Synthesis, Volume I, Erdman A.G and George N. Sandor G. N.
- Theory of Machines and Mechanisms, Shigley J. E and Uicker J. J.
- Vibrations in Rotating Machinery, Professional Engineering Publishers, Wiley

COURSE OUTLINE

- Rotary balancing
- Mechanisms: Definitions and terminology, velocities using analytical methods, instantaneous centre method and relative velocity method; acceleration in mechanisms
- Transmission of rotational motion: Gear trains (including epicyclic trains), belt, rope and chain drives
- Free and forced vibrations of one-degree-of-freedom systems with and without viscous damping
- Introduction to torsional vibration

BALANCING

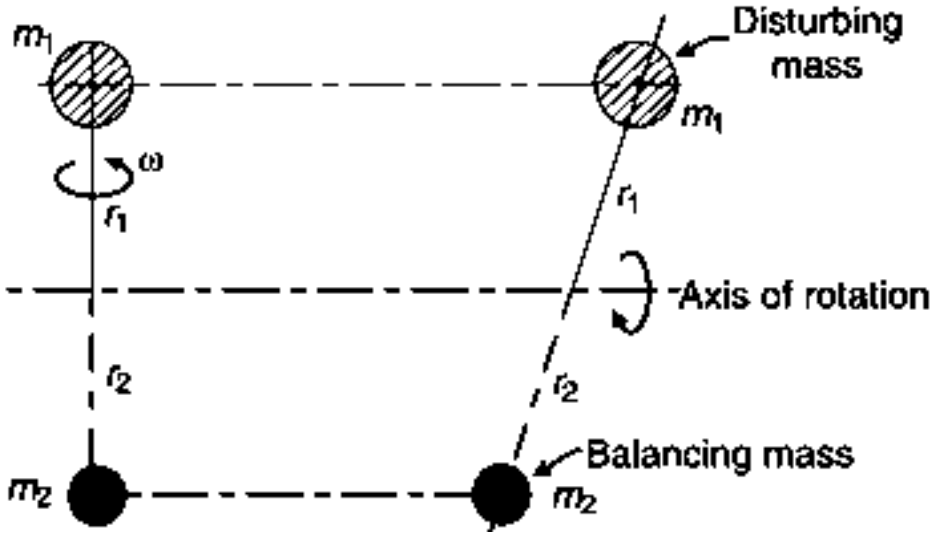
- WHAT IS BALANCING ?
- According to ISO definition, balancing is procedure by which the mass distribution of a rotor is checked and if necessary, adjusted in order to ensure that the vibration of the journals and/or force on the bearings at a frequency corresponding to the service speed is within specified limits.
- Everything that rotates needs to be in a state of balance to ensure smooth running when in operation.
- Precision balancing is essential to the manufacture of rotating equipment and to the repair and renovation of installed machines. As machine speeds increase, the effects of unbalance become more detrimental.
- Modern technology allows for accurate balancing to be performed both in the field and in the workshop.

- EFFECTS OF UNBALANCE
- There are many documented “good consequences” and benefits associated with a well balanced, smooth running rotor. Included among these are:
- **Minimize vibration**
- **Minimize noise**
- **Minimize structural stress**
- **Minimize operator fatigue and annoyance**
- **Increase machine life**
- **Increase bearing life**
- **Increase product quality**
- **Increase personnel safety**
- **Increase productivity**
- **Lower operating costs**

BALANCING

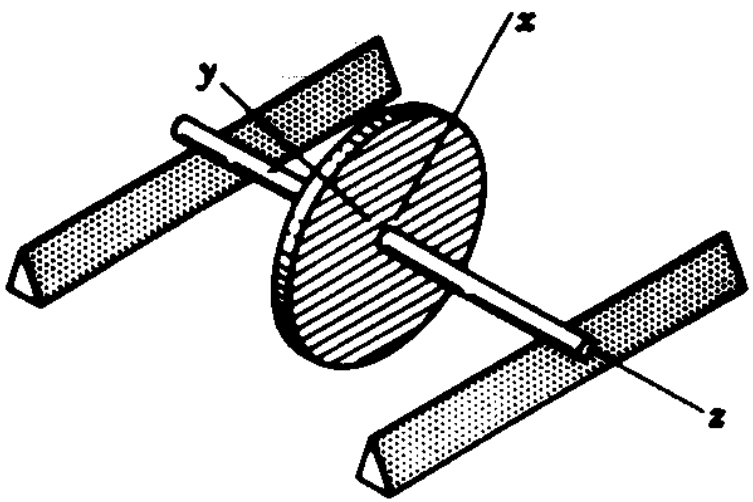
- ✓Balancing of rotating engines
  - Same plane (Static)
  - Different planes (Dynamic)
- ✓Balancing of reciprocating engines
  - Inline
  - Radial
    - V-Engines
    - W-Engines etc.

Balancing of a Single Rotating Mass By a Single Mass Rotating in the Same Plane



This centrifugal force acts radially outwards and thus produces bending moment on the shaft. In order to counteract the effect of this force, a balancing mass ($m2$) may be attached in the same plane of rotation as that of disturbing mass ($m1$) such that the centrifugal forces due to the two masses are equal and opposite.

Balancing of single rotating mass by a single mass on the same plane

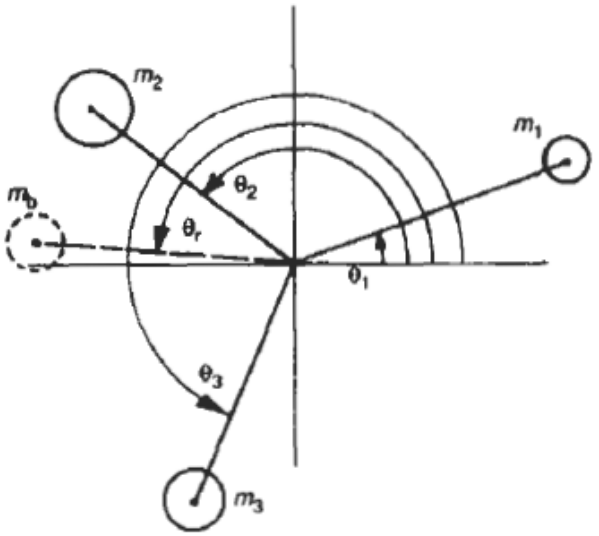


Experiment to determine whether the disc is balanced

- Roll the disc gently by hand and allow it to coast until it comes to rest
- Mark with chalk the lowest point
- Repeat these steps four or five times
- if all the marks coincide it means the disc is not balanced, otherwise is balanced

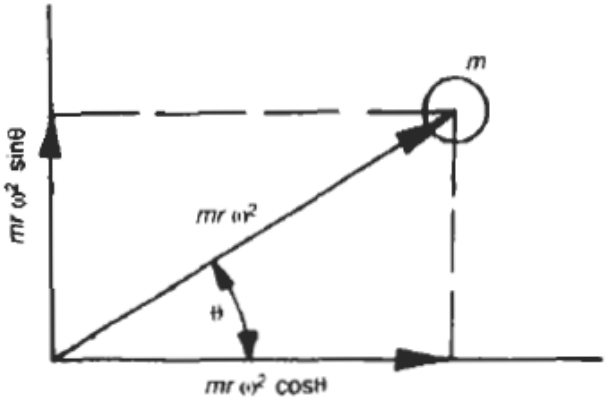
Balancing of rotating masses
Same plane (Static)

Balancing of rotating components is of extreme importance, especially in the *case* of high-speed machinery. Lack of balance may be due to masses in one plane or masses in different planes some distance apart. The method of balancing analytically is given.



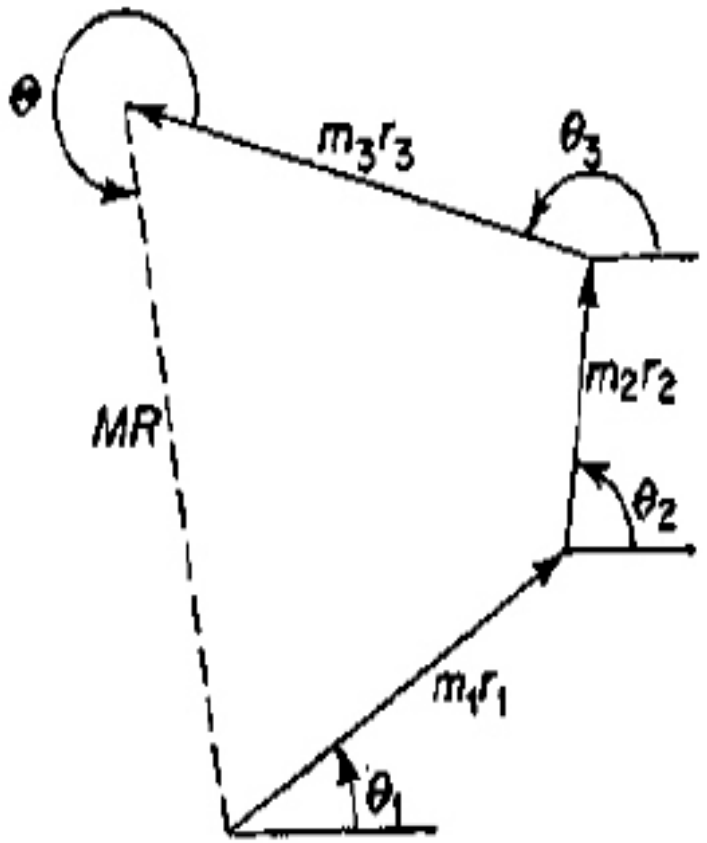
The forces are: $m_1\omega^2 r_1$, $m_2\omega^2 r_2$, etc. These are resolved into vertical and horizontal components:

$$F_V = m_1\omega^2 r_1 sin\theta_1 + m_2\omega^2 r_2 sin\theta_2 + \cdots$$
$$F_H = m_1\omega^2 r_1 cos\theta_1 + m_2\omega^2 r_2 cos\theta_2 + \cdots$$



Resultant force, $F = \sqrt{{F_V}^2 + {F_H}^2}$ at an angle to horizontal axis $\theta = \tan^{-1}\left(\frac{F_V}{F_H}\right)$

Graphical Method



Balancing of rotating masses
Same plane (Static)

EXAMPLE1

Four masses m_1, m_2, m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between *successive* masses are 45°, 75° and 135°. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m.

ANALYTICAL METHOD

$$\sum (mr)_V = \sum_{i=1}^{n} m_i r_i \sin\theta_i = 0$$

$$200 \times 0.2sin0 + 300 \times 0.15sin45 + 240 \times 0.25sin120 + 260 \times 0.3sin255 + 0.2\text{msin}\emptyset = 0$$

$$\text{msin}\emptyset = -42.2 \ \ldots\ldots 1$$

$$\sum (mr)_H = \sum_{i=1}^{n} m_i r_i \cos\theta_i = \mathbf{0}$$

$$200 \times 0.2cos0 + 300 \times 0.15cos45 + 240 \times 0.25cos120 + 260 \times 0.3cos255 + 0.2\text{msin}\emptyset = 0$$

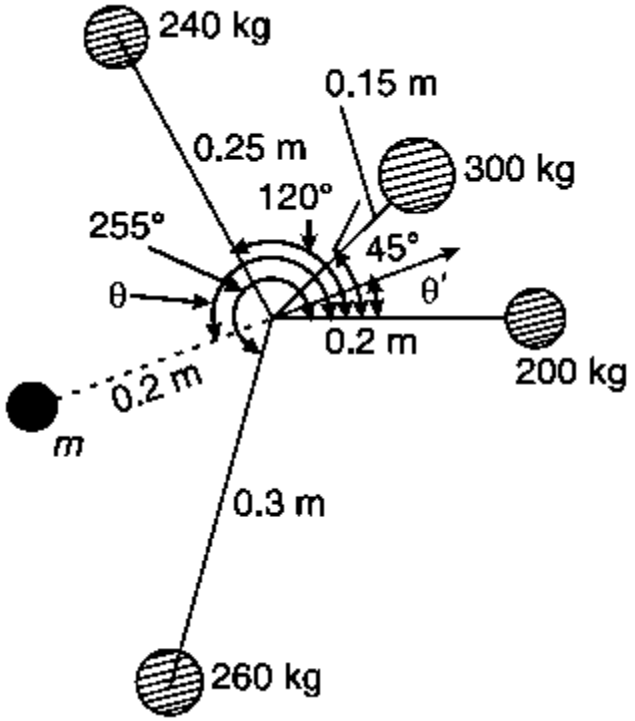
$$\text{m}cos\emptyset = 108.16 \ \ldots\ldots 2$$

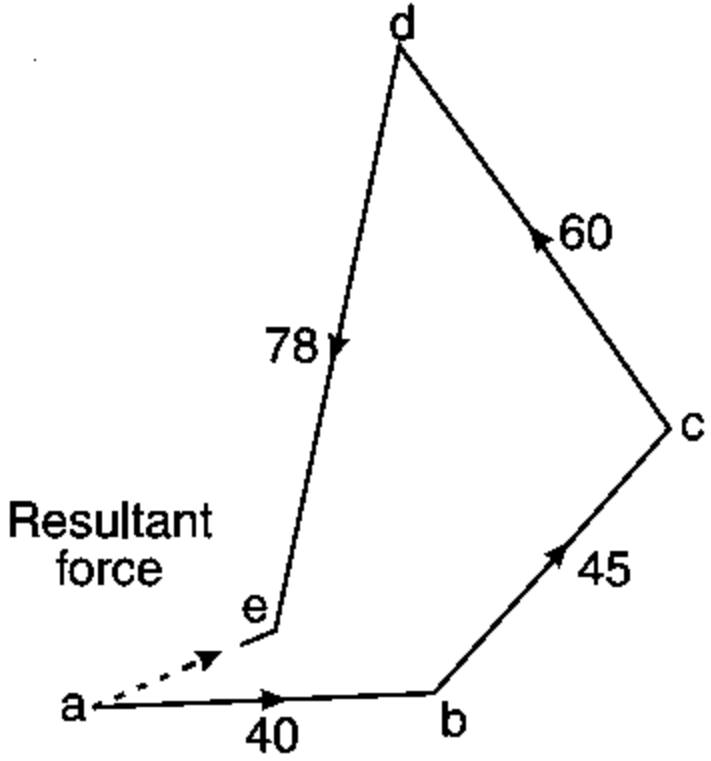
$$\text{m} = \sqrt{(42.2)^2 + (108.16)^2} = -116.1 \text{ kg}$$

$$\theta = \tan^{-1}\left(\frac{8.5}{21.6}\right) = 21.48°$$

$$\phi = \tan^{-1}\left(\frac{8.5}{21.6}\right) = 201.48°$$

Graphical Method





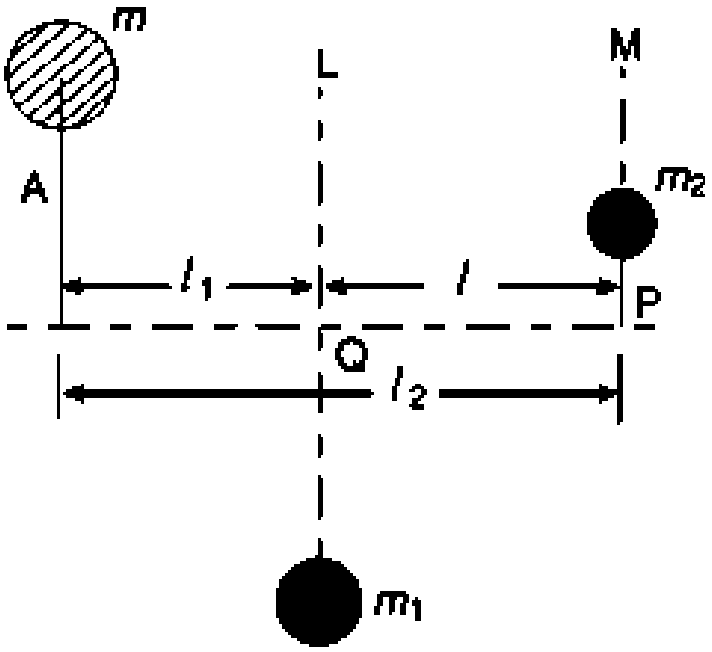
	Mass (kg)	Radius(m)	mr(kgm)
1	200	0.2	40
2	300	0.15	45
3	240	0.25	60
4	260	0.3	78

ae=23 kgm = mr

Unbalance mass, M

M=mr/r = 23/0.2= 115 kg

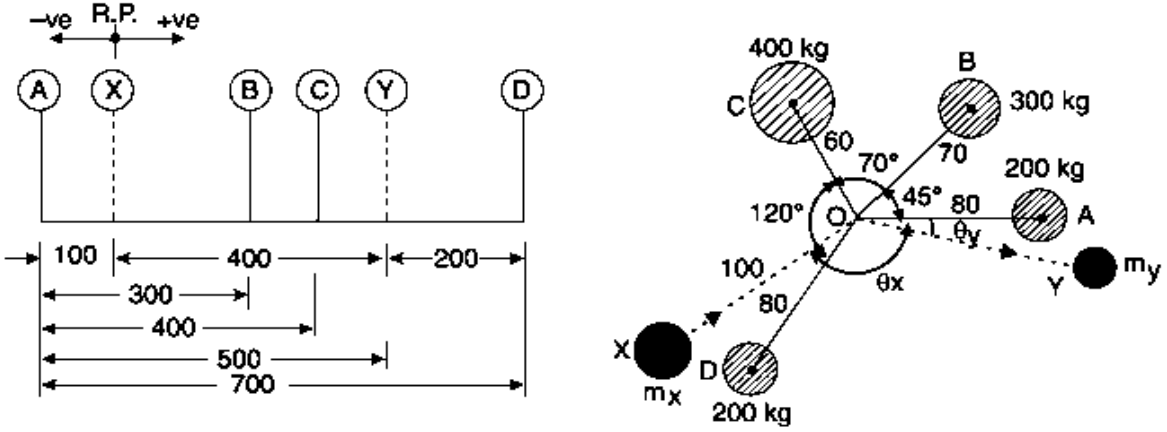
When the plane of the disturbing mass lies on one end of the planes of the balancing masses



- In this case, the mass *m* lies in the plane *A* and the balancing masses lie in the planes *L* and *M*. Two conditions must be satisfied in order to balance the system, *i.e*.
- $\sum F = 0$
- $\sum M = 0$

EXAMPLE 2

A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.



All dimensions in mm.

(*a*) Position of planes. (*b*) Angular position of masses.

Solution Example 2
Analytical Method

Plane	Mass(M)-kg	Radius(R)-m	MR-kgm	L from X(R.P)-m	MRL-kgm^2	θ°	MRcosθ°	MRsinθ°	MRL cosθ°	MRL sinθ°
A	200	0.08	16	-0.1	-1.6	0	16	0	-1.60	0
X(R.P)	mx	0.10	0.1mx	0.0	0.0	θx	0.1mx cosθx	0.1mxsinθx	0.00	0
B	300	0.07	21	0.2	4.2	45	14.85	14.85	2.97	2.97
C	400	0.06	24	0.3	7.2	115	-10.14	21.75	-3.04	6.53
Y	my	0.10	0.1my	0.4	0.04my	θy	0.1my cosθy	0.1my sinθy	0.04my cosθy	0.04my cosθy
D	200	0.08	16	0.6	9.6	235	-9.81	-13.11	-5.51	-7.85

$$\sum mrlcos\theta = 0.04\text{mycos}\theta_Y - 7.18 = 0;$$

$$\sum mrlsin\theta = 0.04\text{mysin}\theta_Y + 1.67 = 0;$$

$$\text{mycos}\theta_Y = 179.5;$$

$$\text{mysin}\theta_Y = -41.75;$$

$$my = \sqrt{179.5^2 + 41.75^2} = 184.29\ kg < -13.09°;$$

$$\sum mrcos\theta = 0.1\text{mxcos}\theta_X + 29.48 = 0;$$

$$\sum mrsin\theta = 0.1\text{mxsin}\theta_X - 19.32 = 0;$$

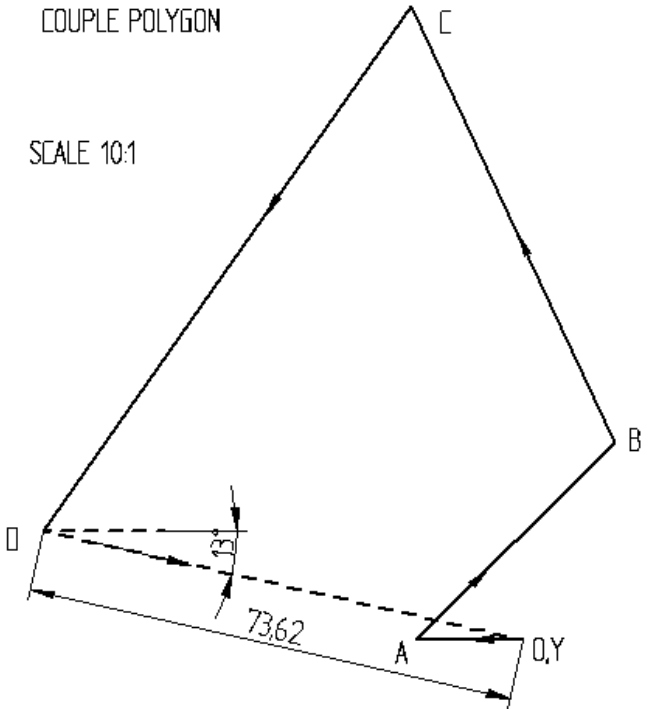
$$\text{mxcos}\theta_x = -294.8;$$

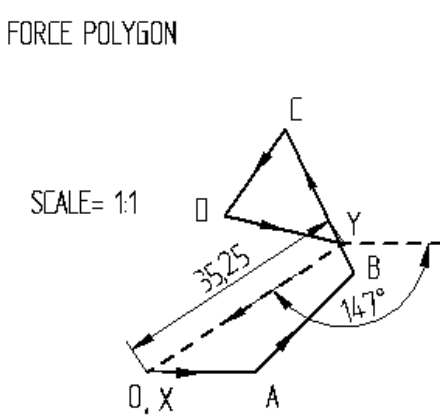
$$\text{mxsin}\theta_X = 193.2;$$

$$my = \sqrt{294.8^2 + 193.2^2} = 352.5\ kg < -146.76°;$$

Solution Example 2
Graphical Method

Plane	Mass(M)-kg	Radius(R)-m	MR-kgm	L from X(R.P)-m	MRL-kgm^2	θ°
A	200	0.08	16	-0.1	-1.6	0
X(R.P)	mx	0.10	0.1mx	0.0	0.0	θx
B	300	0.07	21	0.2	4.2	45
C	400	0.06	24	0.3	7.2	115
Y	my	0.10	0.1my	0.4	0.04my	θy
D	200	0.08	16	0.6	9.6	235





From the Couple Polygon
7.362=0.04my
my=184.05 kg<-13□

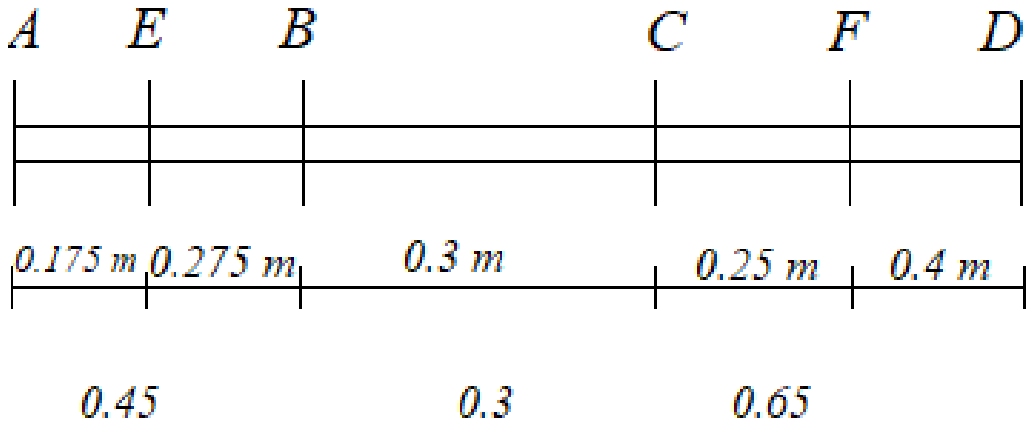
And from the Force Polygon
35.25=0.1mx
mx=352.5 kg<-147□

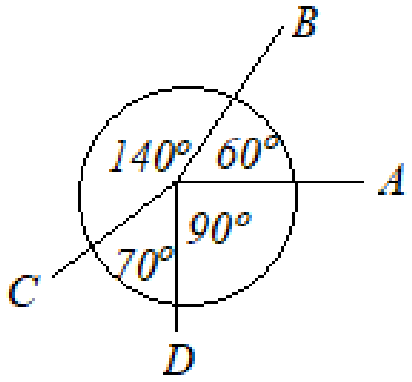
Example 3

A shaft 14 m long carries four eccentric loads A, B, C, and D, spaced at 0.45, 0.75 and 1.4 m from one end. The loads are respectively 7.5, 11, 15 and 6 kg and eccentricities are 36, 48, 54, and 96 mm. The direction of the eccentricities of B, C and D relative to A are 60˚, 200˚ and 270˚. The shaft carried in bearings E and F, which are 0.175 m and 1.0 m from A. E being between A and B.

a) Determine the maximum and minimum vertical forces on bearing F when the shaft rotates at 90 rpm.
b) What is the maximum speed at which the shaft may run to ensure that the vertical component of the load on F is always downwards?

Solution





	Mass (kg)	Radius(mm)	mr(kgm)	L from E (m)	mrl (kg m2)
A	7.5	36	0.270	-0.175	-0.047
E	-		Re	0	0
B	11	48	0.528	0.275	0.145
C	15	54	0.810	0.575	0.466
F	-		Rf	0.825	0.825Rf
D	6	96	0.576	1.225	0.706

Solution

$$\sum_{i=1}^{n} m_i r_i \sin\theta_i = 0$$

$$0.047sin0° + 0.145sin60° + 0.466sin200° + 0.825R_f sin\emptyset + 0.706\sin270° = 0$$

$$R_f sin\emptyset = 0.897 \ldots\ldots 1$$

$$\sum_{i=1}^{n} m_i r_i \cos\theta_i = 0$$

$$0.047cos0° + 0.145cos60° + 0.466cos200° + 0.825R_f cos\emptyset + 0.706cos270° = 0$$

$$R_f cos\emptyset = 0.5 \ldots\ldots 2$$

$$R_f = \sqrt{(0.5)^2 + (0.897)^2} = 1.027 \text{ kgm}$$

$$\phi = \tan^{-1}\left(\frac{0.897}{0.5}\right) = 60.86°$$

Dynamic reaction at F =

$$F_{Df} = R_f \omega^2 = 1.027\left(\frac{2\pi}{60} \times 90\right)^2 = 91.22\ N$$

Solution

Static reaction at F

$$\sum F = 0$$

0.175x 7.5g+0.825xF_{Sf}=0.275x11g+0.575x15g+1.225x6g

$$F_{Sf} = 210.32\ N$$

Maximum vertical force 210.32+91.22=311.54 N

Minimum vertical force 210.32-91.22=119.09 N

For the vertical component of the force to be always downwards

$$R_f \omega^2 \leq 210.32$$

$$1.027\left(\frac{2\pi}{60} \times N\right)^2 \leq 210.32$$

N=136.66 rpm